High reliability series

Serial EEPROMs

ROHM's Serial EEPROMs represent the utmost in reliability.
The lineup is available in the three standard interfaces for compatibility with virtually any application.

New Series

Low 1.7V operation

New product

A new addition to ROHM's lineup of high reliability serial EEPROMs with double cell and double reset technologies are 1.7V low voltage drive models integrating a new, high efficiency circuit in an ultra-compact package. Optimized for battery driven devices and compact modules requiring the lowest operating voltages possible.

BR24S□□□series BR25S□□□series

High reliability EEPROM

Low voltage operation from Vcc=1.7V

High-speed operation

BR24S□□□series
(I²C BUS type) : 400kHz(Vcc=1.7V)
BR25S□□□series
(SPI BUS type) : 20MHz(Vcc=4.5V)

Ultra-compact package

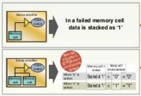
New
2.0mm×3.0mm : VSON008X2030

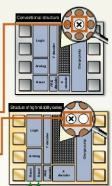
Actual size
VSON008X2030

The reasons behind ROHM's unparalleled reliability.

Double cell

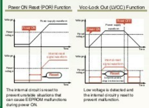
Data is written to the EEPROM by passing electrons through a tunnel-oxide film. This process places a significant amount of stress on the film, possibly leading to erosion and, subsequently memory bit failures that can prevent data from being written to the cells. To avoid this, ROHM utilizes a unique double cell structure that allocates two cells for each bit of data. Therefore, if one cell malfunctions the other will continue to work independently.





Double reset

ICs are generally robust against power interruptions (i.e. ON/OFF) and will normally recover. EEPROMs, on the other hand, are more susceptible, and may malfunction due to unstable power conditions. In order to combat this ROHM integrates two protection circuits, a Power ON Reset and a Vcc Lockout circuit, that detect power interruptions and low voltage conditions during power ON/OFF and resets the internal circuitry, preventing writing errors.



I²C BUS

BR24□□□□ family
- 2-wire communication
- Simple system
- 8bit operation

Series	Supply voltage / Operating temperature	Features
BR24L□□series	Supply voltage 1.8V to 5.5V, Operating temperature −40°C to +85°C	DOUBLE CELL, DOUBLE RESET, 1.8V~5.5V, −40~+85°C, WRITE 5ms, PAGE WRITE
New BR24S□□□series	Supply voltage 1.7V to 5.5V, Operating temperature −40°C to +85°C	DOUBLE CELL, DOUBLE RESET, 1.7V~5.5V, −40~+85°C, WRITE 5ms, PAGE WRITE, 400kHz, 4kV
*BR24H□□series	Supply voltage 2.5V to 5.5V, Operating temperature −40°C to +125°C	DOUBLE CELL, DOUBLE RESET, 2.5V~5.5V, −40~+125°C, WRITE 5ms, PAGE WRITE, 6kV

*: Under development

SPI BUS

BR25□□□□ family
- High-speed communication 20MHz
- Data-protection function (software / hardware area designation)
- 8bit operation

Series	Supply voltage / Operating temperature	Features
BR25L□□0series	Supply voltage 1.8V to 5.5V, Operating temperature −40°C to +85°C	DOUBLE CELL, DOUBLE RESET, 1.8V~5.5V, −40~+85°C, WRITE 5ms, PAGE WRITE, STATUS REGISTER
New BR25S□□□series	Supply voltage 1.7V to 5.5V, Operating temperature −40°C to +85°C	DOUBLE CELL, DOUBLE RESET, 1.7V~5.5V, −40~+85°C, WRITE 5ms, PAGE WRITE, STATUS REGISTER, 20MHz, 4kV
BR25H□□0series	Supply voltage 2.5V to 5.5V, Operating temperature −40°C to +125°C	DOUBLE CELL, DOUBLE RESET, 2.5V~5.5V, −40~+125°C, WRITE 5ms, PAGE WRITE, STATUS REGISTER, 6kV

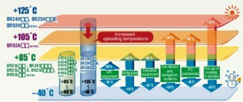
Microwire BUS

BR93□□□ family
- High speed communication 2MHz
- Origin serial EEPROMs
- 16bit operation

Series	Supply voltage / Operating temperature	Features
BR93L□□series	Supply voltage 1.8V to 5.5V, Operating temperature −40°C to +85°C	DOUBLE CELL, DOUBLE RESET, 1.8V~5.5V, −40~+85°C, WRITE 5ms, WRITE ALL, READY BUSY
BR93A□□series	Supply voltage 2.5V to 5.5V, Operating temperature −40°C to +105°C	DOUBLE CELL, DOUBLE RESET, 2.5V~5.5V, −40~+105°C, WRITE 5ms, WRITE ALL, READY BUSY
BR93H□□series	Supply voltage 2.7V to 5.5V, Operating temperature −40°C to +125°C	DOUBLE CELL, DOUBLE RESET, 2.7V~5.5V, −40~+125°C, WRITE 10ms, WRITE ALL, READY BUSY, 8kV

+125°C operation

Sophisticated circuit design and process technology are utilized to widen the operating temperature range, from −40°C to +125°C. This makes them suitable for a wide range of applications, particularly those requiring a greater degree of reliability.



Explanation of Icons

Operating temperature range · Double cell · Double reset · Supply voltage range · Operating temperature range · Write cycle time · Equipped with Write All function · Ready/Busy status display function included · Equipped with Page Write function · Built-in status register · Maximum operating frequency · 4/6/8kV ESD Resistance

General Purpose ICs
Video and Imaging ICs
Audio ICs
IT Equipment ICs
Consumer Product ICs
Automotive ICs
Discrete Semiconductors
Passive Components
Optoelectronics
Modules

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